Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/21/2009
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Telomeric proteins in the radiation/DNA damage response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	susan.bailey@colostate.edu	**Fax:**	FY 970-491-7742
PI Organization Type:	UNIVERSITY	**Phone:**	970-491-2944
Organization Name:	Colorado State University		
PI Address 1:	Environmental and Radiological Health Sciences		
PI Address 2:	1618 Campus Delivery		
PI Web Page:			
City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
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Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
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COI Name (Institution):	Liber, Howard (Colorado State University)		
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	This research addresses Radiation Health Bioastronautics Roadmap (BR) Carcinogenesis Questions 28c, d and j, and is aimed at advancing current knowledge and mechanistic understanding of the DNA damage processing that underlies radiation effects in general, and space radiation in particular. The goal of the studies proposed here is to provide information that will aid identification of susceptible individuals, ultimately improving risk prediction for radiation carcinogenesis in astronauts exposed to high energy charged particles in space. Summary: Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity-they are both DNA ends, the very substrate both		

Task Description:	telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends. Recent and intriguing studies suggesting that the classic telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around-what are telomere proteins doing at DSB ends? Here, we propose to address this fascinating question by investigating the role(s) telomeric proteins play in ionizing radiation (IR)-induced DNA damage response/processing/repair. Our approach assesses mutation frequency and spectra at the molecular level, as well as extends the analyses to the cytogenetic level. Our preliminary studies lend support to the view that telomeric proteins aren't just for telomeres anymore. Increasing our understanding of previously unrecognized relationships between the radiation/DNA damage response and telomere function will facilitate predicting the carcinogenic risk of exposure to radiation in general, and to astronauts exposed to HZE particles in particular. Ultimately, we foresee utilizing telomere instability/dysfunction as a useful biomarker of underlying repair deficiency in order to identify individual susceptibility to radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of interdependencies between DNA repair and telomeres will ultimately lead to identification and characterization of previously unappreciated telomere dysfunction, which in turn will provide a useful indication of an underlying repair defect. Therefore, telomere instability may ultimately serve as a sensitive biomarker, identifying individual susceptibility to radiation carcinogenesis in general and individual susceptibility to HZE exposure in particular.
Task Progress:	Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity – they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends [1]. Recent and intriguing studies suggesting that the “classic” telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around – what are telomere proteins doing at DSB ends? We have recently shown that ionizing radiation (IR)-induced DSBs are insufficient to initiate TRF2 recruitment to damage sites [2]. We are now working to further address this fascinating question by investigating the role(s) telomeric proteins play in IR-induced DNA damage response/processing/repair. Our overall aim is to study the effects of depleting expression of key telomeric binding proteins on DNA damage response/repair and its genotypic and phenotypic consequences. The telomeric DNA-binding proteins TRF1 and TRF2 have multiple interacting binding partners, including TIN2, POT1 and tankyrase1. Many interactions between these players (and others) have been reported. Interestingly, TIN2 has recently been shown to have novel extra-telomeric functions in nuclear organization. Telomerase (hTERT) is also known to interact with many telomeric as well as non-telomeric proteins (including DNA repair proteins), and extra-curricular/pro-tumorigenic activities of telomerase independent of its classical role in telomere length maintenance - including enhanced repair - have been proposed, but remains an open issue. We plan to knockdown expression of these critical telomeric proteins by siRNA. DNA damage will then be induced by exposure to IR (low LET gamma-rays or HZE). We then use a powerful combination of mutational analyses and cytogenetic approaches to examine the phenotypic and genotypic consequences of the deficiencies on repair endpoints. We have found that small interfering RNA (siRNA) knockdown of TRF2 results in a significant decrease in the level of gamma-ray induced mutation at the heterozygous thymidine kinase (tk) locus in human lymphoblastoid cells. Conversely, knockdown of the Protection of Telomeres (POT1) protein results in an increase in radiation mutagenesis. Additional evidence that POT1 plays a critical role in DNA damage response/repair and tumorigenesis is presented in our manuscript demonstrating that mouse Pot1 deficiency promotes telomeric recombination, chromosomal instability and malignant transformation in vivo [3]. Intrigued by the dynamics of the seemingly contradictory yet integrated cellular responses to the requisites of preserving telomere integrity while also efficiently repairing damaged DNA, we investigated roles of the telomeric PARP tankyrase 1 in telomere function and the DNA damage response. Utilizing siRNA knockdown of tankyrase 1 in human cells, we found that its deficiency did not lead to loss of chromosomal end protection per se, but instead resulted in increased levels of telomeric recombination, specifically telomere sister chromatid exchange (T-SCE). Consistent with defective DNA damage response, we also observed increased sensitivity to ionizing radiation-induced cell killing, mutagenesis and chromosome aberrations; similar levels were observed following both gamma-ray and HZE (1 GeV 56Fe) exposures. Most unexpected however, was the finding that tankyrase 1 siRNA knockdown also led to rapid reduction of DNA-PKcs protein levels, while Ku86 and ATM levels remained unchanged. DNA-PKcs mRNA levels were also unaffected. To explore underlying mechanisms, we demonstrated that depletion of tankyrase 1 results in rapid DNA-PKcs proteasome-mediated degradation, likely explaining the associated radiosensitivity phenotype. While reciprocal interactions between PARP1 and DNA-PKcs have been reported, we provide the first evidence to our knowledge for tankyrase 1 – DNA-PKcs protein-protein interaction, and in so doing reveal a new aspect of DNA-PKcs regulation. This work is currently under consideration for publication in PLoS Genetics: Tankyrase 1, telomeric recombination, DNA repair and DNA-PKcs stability by Junqing Zhou, Rupa R. Idate, Ryan Dregalla, Christine L.R. Battaglia, Howard L. Liber and Susan M. Bailey. References [1] Williams ES, Klingler R, Ponnaiya B, Hardt T, Schrock E, Lees-Miller SP, Meek K, Ullrich RL, Bailey SM. Telomere dysfunction and DNA-PKcs deficiency: characterization and consequence. Cancer Res 2009;69 (5):2100-7. [2] Williams ES, Stap J, Essers J, Ponnaiya B, Luijsterburg MS, Krawczyk PM, Ullrich RL, Aten JA, Bailey SM. DNA double-strand breaks are not sufficient to initiate recruitment of TRF2. Nat Genet 2007;39 (6):696-8; author reply 8-9. [3] Wu L, Multani AS, He H, Cosme-Blanco W, Deng Y, Deng JM, Bachilo O, Pathak S, Tahara H, Bailey SM, Deng Y, Behringer RR, Chang S. Pot1 deficiency initiates DNA damage checkpoint activation and aberrant homologous recombination at telomeres. Cell 2006;126 (1):49-62.

Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Williams ES, Klingler R, Ponnaiya B, Hardt T, Schrock E, Lees-Miller SP, Meek K, Ullrich RL, Bailey SM. "Telomere dysfunction and DNA-PKcs deficiency: characterization and consequence." Cancer Res. 2009 Mar 1;69(5):2100-7. PubMed PMID: 19244120 , Mar-2009